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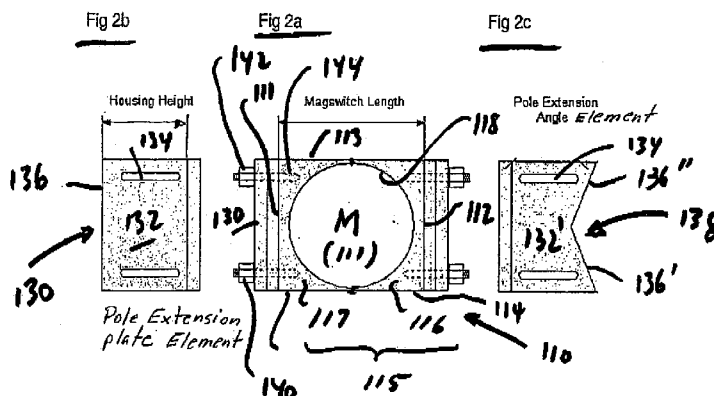
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(54) Title: SWITCHABLE MAGNETIC IMPLEMENT



(57) Abstract: Described are switchable magnetic implements for use a work piece or tool holders and as jigs for machining or welding operations, the implement having at least two permanent magnets received within a housing having or comprised of at least two main pole pieces, one or both of the magnets being movable to selectively generate an external magnetic field and a magnetic flux path through the pole pieces when in contact with a ferromagnetic work piece, which is characterised by including at least two pole extension members which are respectively secured, preferably in exchangeable manner, to the associated one of the main pole pieces, and in that the pole extension members have at least two engagement surfaces adapted to be brought into contact with two work pieces, the engagement surfaces being contoured to at least partially match the contours of differently shaped work pieces, eg provide a curved or non-rectilinear engagement or contact location as well as rectilinear contact / engagement surfaces at the same implement.



SWITCHABLE MAGNETIC IMPLEMENT

FIELD OF THE INVENTION

The present invention relates to switchable magnetic devices used to secure work pieces for and during metal working applications (such as welding, machining, joining, etc) or to serve as holders for ancillary tools used during such metal working or generally as a tool holder.

BACKGROUND TO THE INVENTION

The present invention was conceived to provide additional versatility and application fields for switchable permanent magnetic devices, in particular such as described in Patent documents AU753496 (patent application PCT/AU00/01505) and AU2002325077 B1. The entire content of these documents is incorporated herein by way of shorthand cross-reference to these documents.

Briefly, and having reference to figure 1, which illustrates a switchable permanent magnet device as per AU753496, the magnetic device 10 essentially consist of two permanent magnets 11, 12 located within a housing 15 consisting of two pole pieces 16, 17, and an actuator handle 19.

Both magnets 11, 12 are disc or cylindrically shaped magnets which are diametrically magnetised, ie the north pole and the south pole of the magnets are located on opposite sides of a vertical plane that extends over the height of the magnets along a diameter of an upper face of the magnets. Both magnets 11, 12 are rare-earth type magnets, for example neodymium-iron-boron magnets, and are received in stacked, face to face juxtaposition within an open-ended cylindrical cavity of housing 15 comprised of two, low-magnetic reluctance ferromagnetic pole pieces 16, 17 which have a height (in stacking direction of magnets 11, 12) just larger than the height of the stacked magnets 11, 12. The wall thickness of pole pieces 16, 17 is not uniform and the external contour (as per top or bottom plan view) is chosen to maximise the external magnetic field strength which the device is able to provide when switched on as explained below. The pole pieces 16, 17 are magnetically separated at the diametrically opposed joint edges 18a and 18 b; for further details refer to the mentioned prior art documents.

Magnets 11, 12 are received within housing 15 such that lower magnet 11 is prevented from rotating, in a fixed position, whilst upper magnet 12, however, is able to be rotated about the stacking axis (longitudinal axis of device 10) by means of a handle or lever 19 coupled / secured to upper magnet 12. A top cover 20 seals the top of the housing 15 whilst permitting an axle of handle 19 to extend there through and engage top magnet 11; a suitable sealing member assists in forming a water tight and dust proof seal around that opening. The lower magnet 12, which is fixed against movement, provides the lower, sealed cover at the other end of the device 10.

To switch the device 10 between 'on' and 'off' positions in which the device 10 exhibits a strong external and weak (or no) external magnetic field, respectively, the upper magnet 11 requires rotation by 180 degrees from a position in which the respective north and south poles of the stacked magnets 11, 12 are aligned with one another (as viewed in top or bottom plan view), into a rotational position in which the north pole of the top magnet 11 substantially overlies the south pole of bottom magnet 12; it follows that the south pole of top magnet 11 substantially overlies the north pole of bottom magnet 12. In this arrangement, the first and second magnets act as an internal active magnetic shunt and as a result the external magnetic field strength from the device is quite low (and ideally not present).

In the 'on position', the respective magnetic field lines of the upper and lower magnets 11, 12 are superimposed, and the device can be firmly attached to ferromagnetic surfaces by bringing the latter into contact with both pole pieces 16, 17 to provide a closed magnetic circuit. Thus it is to be noted that the passive poles are important in assisting the magnetic functionality. In the embodiment of figure 1, the device would have a working face at the lower end ie at the bottom magnet (which is slightly recessed from the lower most ends of the pole pieces) by means of which the device attaches to a work piece.

A need exists with such devices to provide for improved, work piece specific configuration that facilitate the creation of a closed magnetic flux path at the interface device – work piece. The aim is to ensure that the work piece can be secured in safe and reliable manner to the magnetic device. It would be possible to shape the lower engagement surfaces of the pole pieces of the device of figure

1 to cater for differently shaped work pieces; this however means that such device lacks versatility as regards other work piece engagement configurations.

It would be also advantageous to be able to secure two or more work pieces in fixed spatial relationship to one another using a single magnetic device (or multiple devices at different locations of the same two work pieces. That is, in one application, such device would act as a joining element in its own right, thereby allowing ferromagnetic plates or elements (work pieces) to be joined in removable manner to one another. In another application, the device would provide a jig functionality thereby to facilitate operations that permanently join the work pieces together through other means, like brazing and welding in particular.

Beyond serving as a temporary joining element or jig, another object would be to provide additional functionality which enables attachment of other tools to the magnetic tool or jig, eg to serve as a tool holder or similar.

SUMMARY OF THE INVENTION

In accordance with a broad aspect of the present invention, there is provided a magnetic implement comprising (i) a switchable magnetic device having at least two permanent magnets received within a housing having or comprised of at least two main pole pieces, one or both of the magnets being movable to selectively generate an external magnetic field and a magnetic flux path through the pole pieces, and (ii) at least two pole extension members which are respectively secured, preferably in exchangeable and removable fashion, to an associated one of the main pole pieces, and wherein the pole extension members each have at least two engagement surfaces adapted to be brought into selective contact with one or more work pieces, the engagement surfaces being contoured differently and so to at least partially match the contour of the work piece at an engagement or contact location.

The contour of the at least two engagement surfaces of the pole extension members need not be identical to one another, or identical to that of the work piece. Relevant is that the contact or engagement surfaces that provide the interface between pole extension members and a work piece, are optimised in order to provide an improved flux transfer functionality between the active magnetic material (permanent magnets) through the pole pieces (passive

magnetic material) and pole extension members of the implement, and one or more work pieces.

The expression 'implement' is here used to denote a device that may be a tool or construction element, eg a fastener, coupler, work piece holder, or a jig as used to secure work pieces for further processing, eg such as welding.

Preferably, the two main pole pieces, which form part of the magnetic device or unit, will have each an attachment zone that is contoured to match a corresponding attachment zone of the pole extension members, eg provide a form-fitting and preferably a surface plane interface which minimises magnetic flux losses that an air gap may otherwise create at the interface where the pole extension members attach to the main pole pieces.

To ease manufacture without compromising too much on magnetic efficiency, the housing of the magnetic switchable device may be box-like shaped, with a cylindrical cavity for receiving the diametrically polarized cylindrical permanent magnets. That is, where the main pole pieces define the housing, these will exhibit exterior faces that are rectangular or square and plan, which in turn will require the pole extension members to have planar attachment surfaces to provide form-seating attachment of pole extension members to the pole pieces.

The pole extension members will advantageously be secured / fixed to the device housing (ie the main pole pieces) in modular fashion, eg using bolts whose threaded stems engage in corresponding threaded bores in the main pole pieces. This will allow ease of exchange of pole extension members to match specific applications of the implement.

The pole extension members can advantageously be provided each by one or more ferromagnetic plates having a desired minimum thickness, the contour of the plates being such as to define a polygon. The edge or peripheral sides of the polygonal plates provide the engagement surfaces that can be brought into contact with the one or more work pieces in selective manner.

The polygon / exterior contour of the extension member plates will be preferably irregular so as to provide differently angled adjoining sides and/or differently contoured sides. This measure increases versatility and reduces the number of differently shaped pole extension plates that will need to be ready at hand for matching work piece geometries and jig functionality. For example, the

polygon may be a hexagon with two straight sides joining at 90 degrees, a further straight side joining its neighbour at 120 or 135 degrees (inclusive angle), a curved side having a curvature radius to match the curvature of say a steel pipe of 10 to 100 cm; the final two sides of the hexagonal plates may themselves be contoured or profiled in another geometric pattern that matches an external shape of work pieces intended to be joined or used with the magnetic jig or tool.

The magnetic implement can be used as a jig for welding two ferromagnetic plates together at an angle whilst these are securely held at adjoining engagement surfaces of the pole extension members in abutting relationship. Of course, the work pieces need not be secured at adjoining engagement surfaces, but at other locations of the jig.

Advantageously, such welding jig embodiment will have an integrated grounding 'clamp' facility. To this end, a ground cable fixing member / armature of suitable conductive material like copper or the like is fixed in removable manner to an outside of one of the pole extension plate members.

To create a conductive circuit between the plate members and 'divert' electric current flow from having to pass through the ferromagnetic main pole pieces, it will be advantageous to secure a conductive rod, tube or the like in bridging relationship between the two pole extension plates, the material of this bridge member being advantageously selected to provide less resistance to electric current flow than that of the magnetic device pole pieces; aluminium is a preferred material for this purpose.

To enhance understanding of the present invention, two embodiment of the present invention and further features thereof will be described below in non-exhaustive manner with reference to the accompanying figures.

BRIEF DESCRIPTION OF THE DRAWINGS

Figure 1 shows an isometric, exploded view of the components that make up a switchable magnetic device according to the prior art for incorporation (modified) into a magnetic tool or jig in accordance with the present invention;

Figures 2a to 2c show respectively a schematic top plan view and alternative end side elevation views of a magnetic attachment / holder tool in accordance with a first embodiment of the present invention, fig 2 b showing a

first embodiment and fig 2b a second embodiment of pole extension elements of the tool;

Figures 3a and 3b are schematic top plan and end side elevation views of a further alternative embodiment of pole extension members as can be used in the present invention;

Figures 3c and 3d are schematic top plan and end side elevation views of yet a further embodiment of pole extension members as can be used in the present invention;

Figures 4a and 4b show schematic illustrations of two possible positioning orientations of the device of figure 2, depending on the type of pole extension members employed, on a tubular work piece, ie a steel pipe; and

Figures 5a and 5b are perspective views of a welding jig with ground clamp functionality as per a further embodiment of the present invention.

DESCRIPTION OF PREFERRED EMBODIMENT

Figure 1 illustrates a switchable magnetic device 10 for incorporation into the modular switchable magnetic implement in accordance with the present invention. The basic constituents and their mode of operation, but for the differences noted below, have been described in the context of the background of the invention, and reference should be made above, as well as to above identified prior art patent documents, for a more detailed description of the basic make-up of the switchable active component of the implement which preferred embodiments of the present invention relate to.

The device 10 of figure 1 is modified for the purposes of providing a switchable magnetic unit 110 for incorporation into the magnetic mounting or work piece holding tool 100 illustrated in figures 2, 3 and 4, and switchable magnetic unit 110 of the welding jig 200 illustrated in figure 5a, 5b, in that the geometry / shape of the housing 15 (fig. 1) which is made up by two main pole pieces 16, 17 is changed to define a housing 115 (see figures 2a to 2c) of box-like configuration, the pole pieces 116 and 117 defining a cylindrical cavity 118 for housing the cylindrical permanent magnets (shown in Fig 2a only at M (111)) and external, plan surfaces 111, 112, 113 and 114. The switchable units 10 and 110 will be herein below simply referred to as a Magswitch unit, the functionality thereof being the same in all applications, namely to provide the switchable

component by means of which the implement 100 / jig 200 can be secured to and released from a work piece. The length of the core Magswitch 110 is deliberately kept to a minimum to conserve magnetizing force, whilst the housing height will be dictated primarily by the volume of magnetic active material specified to meet operational requirements, eg holding force / magnetic field strength; in effect, the housing pole piece height is somewhat larger than the dimension of the two stacked magnets.

The work piece holder 100 of figs 2, 3 and 4 incorporates a Magswitch unit 110 and two pole extension members 130 secured to opposite sides 111, 112 of the housing 115 in magnetic and form-fitting attachment to pole pieces 116, 117, respectively.

Figure 2b illustrates a first embodiment of a pole extension member 130 comprised of a rectangular ferromagnetic plate 132 with two long holes 134 that pass through the thickness of plate 132. A lower edge of plate 132 provides an inclined but otherwise rectilinear engagement surface 136.

Figure 2c illustrates a second embodiment of a pole extension member 130, which is again a ferromagnetic steel plate 132' with two long holes 134 extending through the plate 132'. However, instead of having a straight lower engagement edge surface, this extension pole member 132' exhibits two converging surface edges 136' and 136'' that make up a pitched (or notched), complex engagement surface 138.

To assemble the holder 100, either two plates 132 or 132' are firmly secured to Magswitch sides 111, 112 using two bolts 140, 142 whose threaded shafts pass through long holes 134 of plate 132 (or 132') and are received in corresponding threaded bores 144 that extend into the pole pieces 116, 117. The amount by which the pole extension members 130 protrude from a lower end of the Magswitch 110 can be adjusted by loosening bolts 140, 142 and sliding the plates 132 (or 132') along their long holes 134 and retightening the bolts 140, 142.

As may be best seen in figures 4a and 4b, selection and use of the appropriate pole extension member plates 130 will be dictated by not only the shape of the work piece, which in figs. 4 is a steel tube, but also by the orientation which the holding tool 100 is to have on the work piece. Accordingly, it will be

understood that it is envisioned by the present invention to provide a plethora of demountable pole extension elements 130 with varying contours at the engagement surface 136, 138, each suited to match or better follow the shape / contour of the work piece at the engagement location. Such contoured engagement surfaces improve attachment capability of the Magswitch with an object that has an uneven or irregular surface contact zone (e.g. pipes).

In another embodiment of the modularly attachable pole extension members, figures 3a and 3b illustrate multi-piece extension members 150 and 160, respectively. In figure 3a, the pole extension member 150 consists of a plurality of thin, stacked ferromagnetic plates 151, which are sandwiched and fixed against sides 111 and 112 of the Magswitch unit housing 115 at different heights, so as to define a staggered or finely stepped lower engagement surface 156. In contrast, the pole extension member 160 of fig 4b consist of a plurality of individual ferromagnetic rods 162 of quadrilateral configuration, which are received in a regular array with transition fitting in an external u-shaped channel housing 164 that itself is bolted to the Magswitch housing 115 as describe previously. This configuration allows individual rods 162 to be displaced to more closely follow the contour of the work piece, to help improve contact to the work piece and to optimise magnetic performance (variation of wall thickness).

Figures 5a and 5b show different isometric views of yet another embodiment of a switchable magnetic implement according to the present invention. The implement is a multi-face welding jig 200 that allows securing two (or more) ferromagnetic plates in fixed spatial relationship to one another, so as to perform machining operations and/or welding / brazing.

Jig 200 incorporates a Magswitch unit 110 as described above, with an actuator knob or handle 119 to effect turning on and turning off of the unit. The pole extension members are here provided by two identical ferromagnetic plate members 220 of specified thickness, having a polygonal outer edge contour. In this embodiment, the pole extension plates are also removably secured to the housing 115 of the Magswitch unit 110 by bolts (not illustrated) although it is also possible to affix the extension plates permanently to housing 115.

In the embodiment illustrated, the plates have a hexagonal outer rim or edge, with three, discrete rectilinear edge surfaces 222, 224, 226 in adjoining

sequence, whereby the included angle between edge surfaces 222 and 224 being 90 degrees and between edge surfaces 224 and 226 being 135 degrees; these surfaces 222, 224 and 226 enable butt or edge welding of ferromagnetic plates when magnetically secured to/by the jig 200. The two edge surface portions 226 and 228 which follow the rectilinear ones, have each a triangular notch or recess 227 and 229 in them, wherein the shape and height of the triangular indentations are different in sections 228 and 230; these partially recessed edge surfaces 228 and 230 can serve to accommodate / receive triangular or cylindrical rod material and secure same in an arrangement that spans across the two pole extension plates 220; the included angle between the adjoining engagement surfaces 228 and 230 can be different to 90 degrees, and might be 120 or 60 degrees. In closing the perimeter of the pole plates 220, reference number 232 identifies a curved engagement edge surface that can be used to abut the jig 200 against cylindrical work pieces, such as steel pipes; the radius of curvature may be set to suit the specific pipe diameter to be secured / welded. One could describe the magnetic jig with its pole extension plates as providing a 'builders square' that enables two work pieces attached to the same jig 200 to be brought together in safe and geometrically aligned / fixed manner to perform the subsequent welding operation.

It will be appreciated that the specific contour of the engagement surfaces 222 to 232 and the number thereof at the extension pole members can be varied to suit, eg match the shape or contour of a specific work piece type, although a multi-purpose tool would have engagement surfaces contoured for the most common work piece geometries / contours, ie rectilinear and curved.

The welding jig 200 of fig. 5 has an additional 'grounding clamp' functionality for electric arc welding applications. To this end, a welding ground cable armature 250 made of copper or another conductive material is located in conductive engagement on the outer side (face) of one of the pole extension plates 220 and secured thereto by means of a welder's bolt 252 that extends through an appropriately sized through-hole in the plate. Bolt 252 is of such length as to span the gap between the facing parallel pole extension plates 220 and protrude with its terminal threaded end from the other side plate 220 where it is secured with a nut (not shown). To maximise electric conductivity along the

grounding path provided by the armature 250 and minimise a current path across the ferromagnetic pole pieces of the Magswitch unit 110, an aluminium or copper tube 254 surrounds the shaft of bolt 252 and provides an additional, direct current bridge between the pole plates 220; tube is maintained in good contact with both plates 220 by appropriate tensioning of bolt 252.

Finally, it will also be noted that the pole extension plates 220 have each a number of through holes. These can serve to fasten additional or alternative elements to the magnetic tool / jig 220.

CLAIMS:

1. Magnetic implement comprising (i) a switchable magnetic device having at least two permanent magnets received within a housing having or comprised of at least two main pole pieces, one or both of the magnets being movable to selectively generate an external magnetic field and a magnetic flux path through the pole pieces when in contact with a work piece, and (ii) at least two pole extension members which are respectively secured, optionally in removable but preferably in exchangeable fashion, to an associated one of the main pole pieces, the pole extension members each having at least two engagement surfaces adapted to be brought into selective contact with one or more work pieces, the engagement surfaces being contoured to at least partially match the contour of the work piece at an engagement or contact location.
2. Switchable magnetic implement according to claim 1, wherein the contours of at least two of the engagement surfaces of the at least two pole extension members are dissimilar to one another.
3. Switchable magnetic implement according to claim 1 or 2, wherein the two main pole pieces have each an attachment zone that is contoured to match a corresponding attachment zone of the pole extension members.
4. Switchable magnetic implement according to claim 3, wherein the housing of the magnetic switchable device is box-like shaped, with a cylindrical cavity for receiving the diametrically polarized cylindrical permanent magnets and planar exterior surfaces for the attachment zones.
5. Switchable magnetic implement according to any one of the preceding claims, wherein pole extension members are each one or more ferromagnetic plates with edge surfaces thereof defining a polygon providing the at least two engagement surfaces.
6. Switchable magnetic implement according to claim 5, wherein the extension member plates have a peripheral contour defining an irregular polygon

so as to provide adjoining engagement surfaces with varying included angles and/or differently contoured sides.

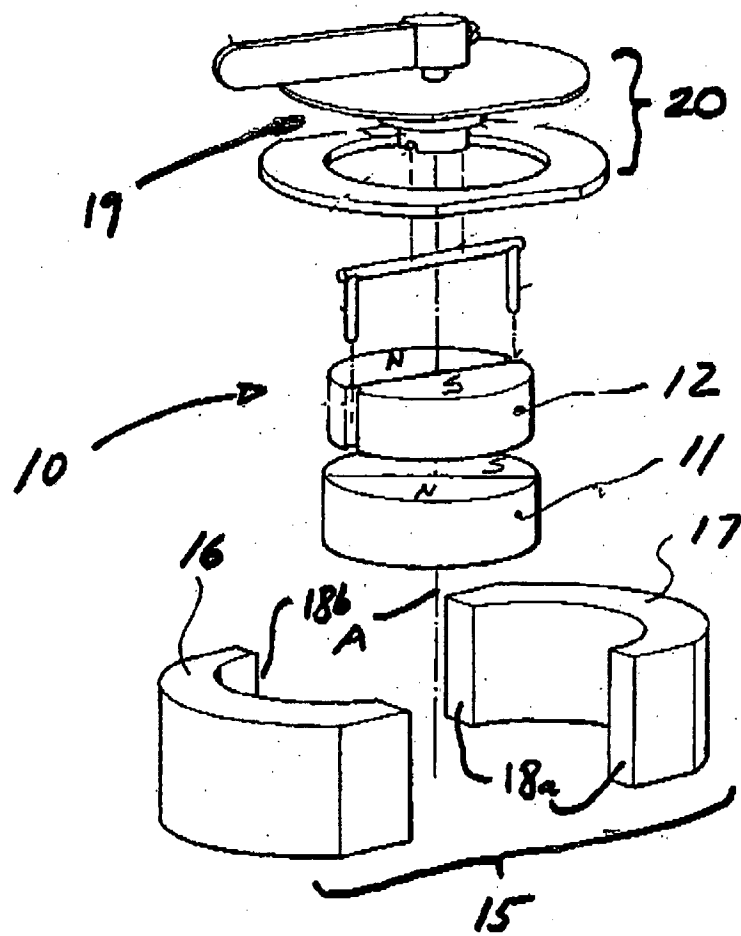
7. Switchable magnetic implement according to claim 5 or 6, wherein the polygon is a hexagon with two straight line sides joining at 90 degrees, a further straight line side joining its neighbour at 120 or 135 degrees inclusive angle, at least one curved side having a curvature radius to match standardised metal pipe dimensions, and at least one side profiled in another geometric pattern that matches at least partially an external shape of a work piece.

8. Switchable magnetic implement according to any one of claims 5 to 7, further including an electrical grounding cable fixing armature of suitable conductive material like copper or the like, fixed in removable manner to an outside of one of the pole extension plate members.

9. Switchable magnetic device according to claim 8, further including a electricity-conductive rod, tube or the likewise conductive member located and fixed in bridging relationship between the two pole extension plate members.

10. Switchable magnetic implement according to any one of the preceding claims, wherein the pole extension members are provided with one or more facilities to anchor one or more tools to the implement.

FIG. 1



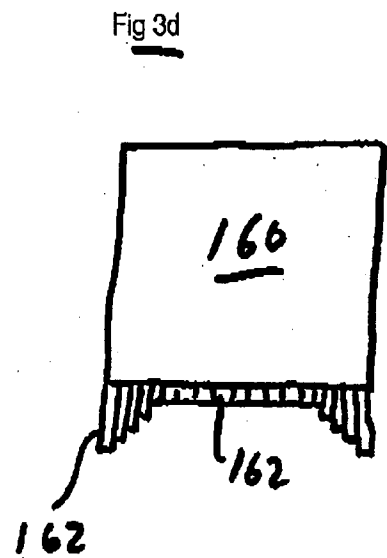
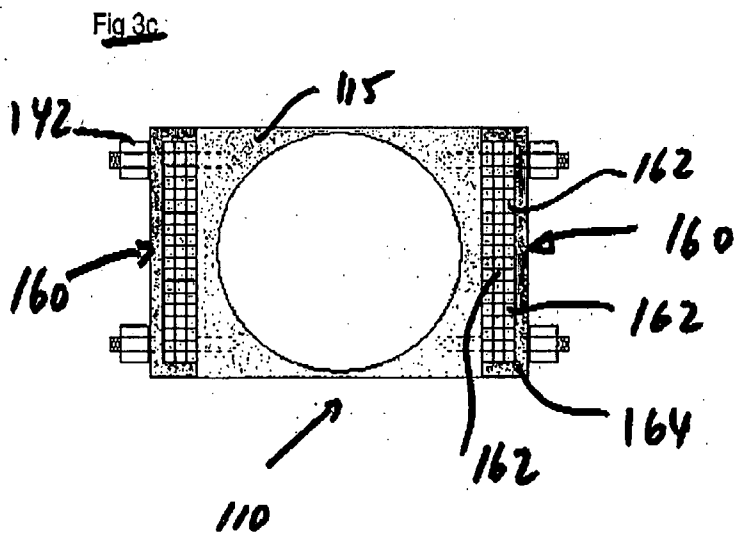
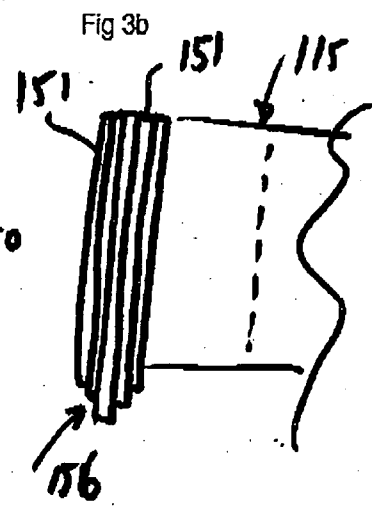
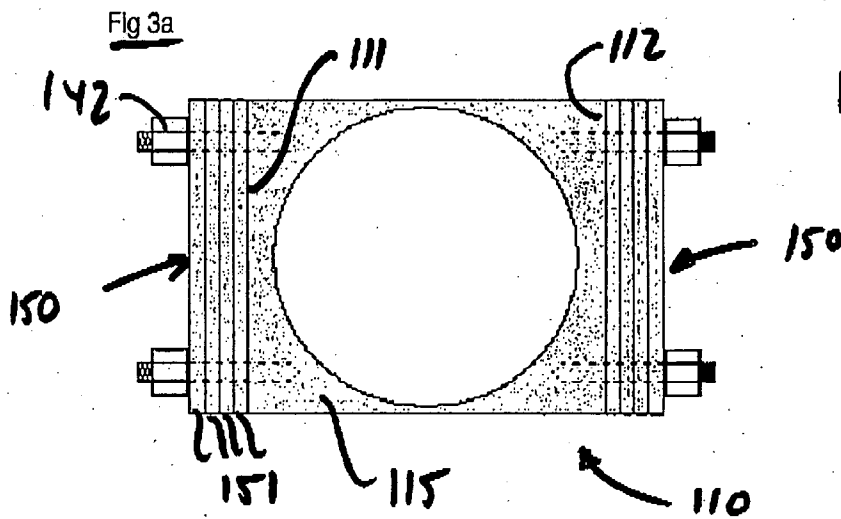
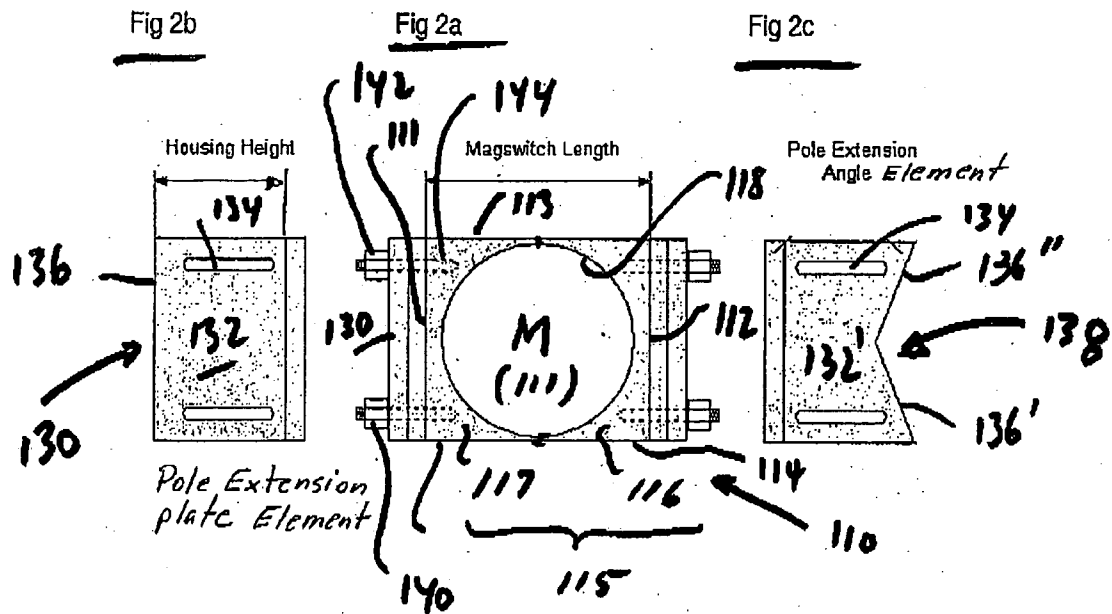


Fig 4a

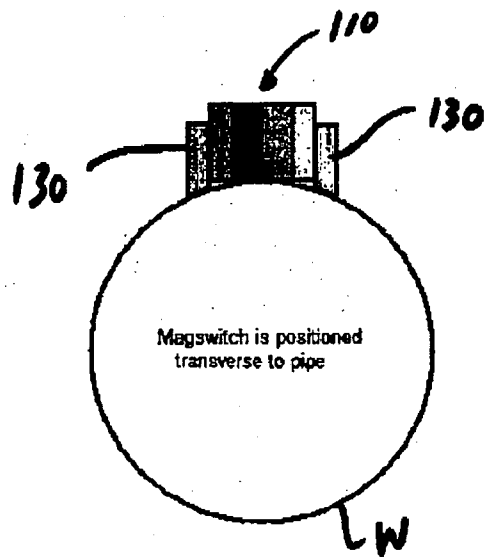


Fig 4b

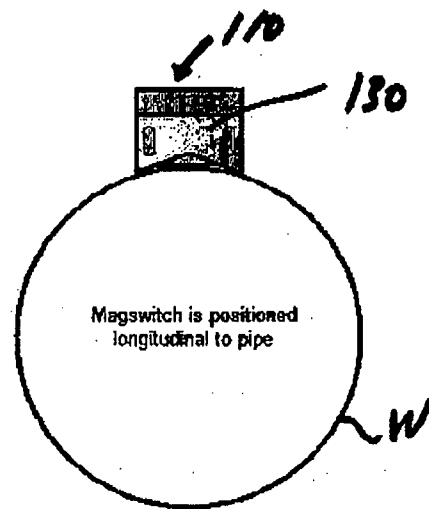


Fig 5a

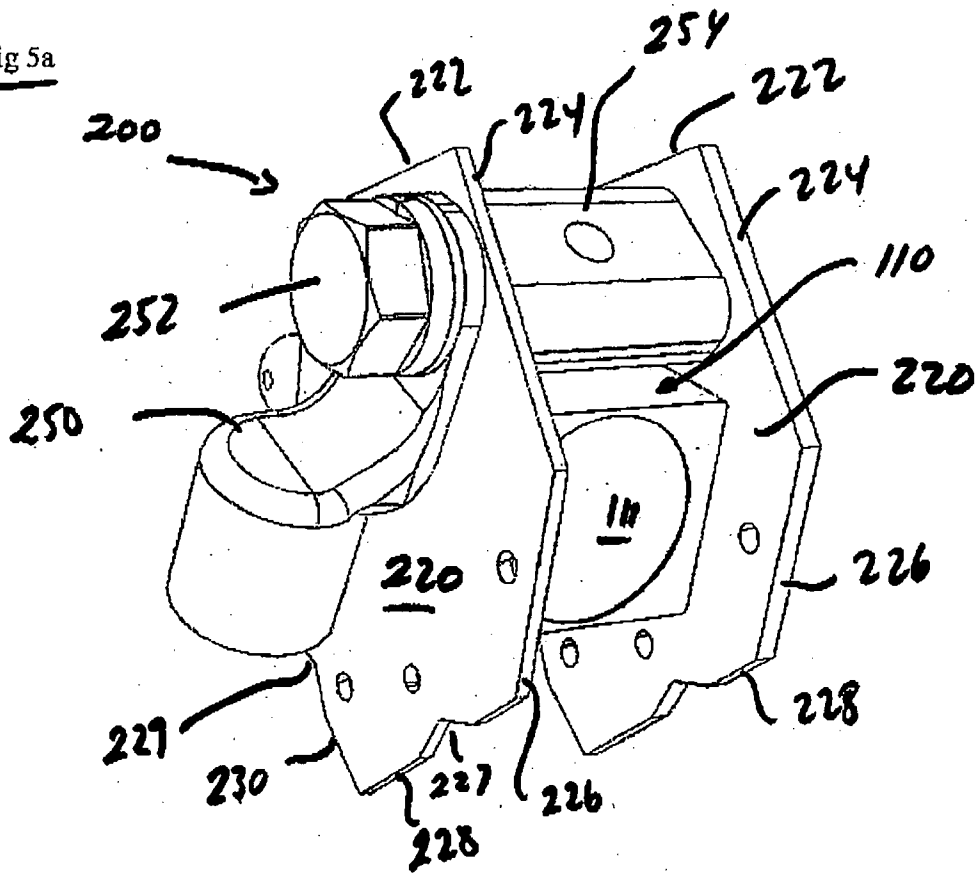
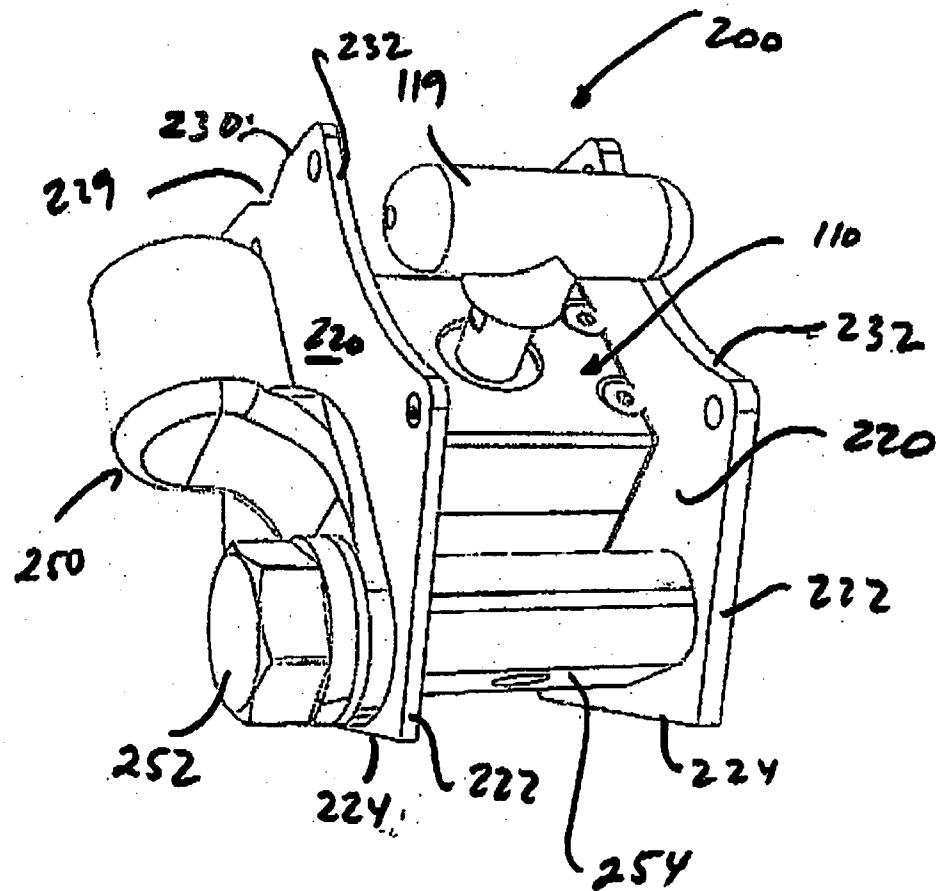


Fig 5b



INTERNATIONAL SEARCH REPORT

International application No.

PCT/AU2010/000666

A. CLASSIFICATION OF SUBJECT MATTER

Int. Cl.

B25B 11/00 (2006.01)**B25B 11/02** (2006.01)

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

WPI, EPODOC: B25B 11/00, 11/02, AND Keywords only (pole+, contour+, shap+, match+, profil+, magnet+, pick_up, hold+, elevat+, hoist+, grip+, field+, flux+, rotat+, mov+) and like terms.

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US 4652845 A (FINKLE) 24 March 1987 Whole document, Figs 1, 2	1-10
A	US 5630634 A (STOWE et al.) 20 May 1997 Whole document (figs 1-4)	1-10
A	US 3363209 A (PEVAR) 9 January 1968 Whole document	1-10



Further documents are listed in the continuation of Box C



See patent family annex

* "A"	Special categories of cited documents: document defining the general state of the art which is not considered to be of particular relevance	"T"	later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"E"	earlier application or patent but published on or after the international filing date	"X"	document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
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"O"	document referring to an oral disclosure, use, exhibition or other means	"&"	document member of the same patent family
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Date of the actual completion of the international search
12 July 2010

Date of mailing of the international search report

22 JUL 2010

Name and mailing address of the ISA/AU

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No.

PCT/AU2010/000666

This Annex lists the known "A" publication level patent family members relating to the patent documents cited in the above-mentioned international search report. The Australian Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

Patent Document Cited in Search Report		Patent Family Member	
US	4652845	AU	61692/86
		FR	2588543
		CN	86107190
		GB	2181897
		DE	3628544
		JP	62202507
US	5630634	NONE	
US	3363209	NONE	

Due to data integration issues this family listing may not include 10 digit Australian applications filed since May 2001.

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